

STAT 412 QZ07 — Question 1

Problem

The average height of females in the freshman class of a certain college has historically been 162.2 centimeters with a standard deviation of 6.7 centimeters. Is there reason to believe that there has been a change in the average height if a random sample of 60 females in the present freshman class has an average height of 164.3 centimeters? Use a P-value in your conclusion. Assume the standard deviation remains the same.

Solution

The claim is that the mean has changed, so this is a two-tailed test:

$$H_0 : \mu = 162.2, \quad H_1 : \mu \neq 162.2.$$

Because the population standard deviation is known, use a z test:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{164.3 - 162.2}{6.7/\sqrt{60}} = 2.43.$$

The two-tailed P-value is

$$P = 2P(Z \geq 2.43) \approx 0.015.$$

Since the P-value is small, reject H_0 for significance levels greater than 0.015.

Final Answer

$H_0 : \mu = 162.2$, $H_1 : \mu \neq 162.2$; $z = 2.43$; P-value = 0.015. Reject H_0 . There is sufficient evidence that the average height has changed.